```sql
-- ===========================================
-- DATABASE CREATION
-- Tables:
-- Departments, Employees, Customers,
-- Products, Orders, Sales, MonthlySales, SalesTeam
-- ===========================================
CREATE DATABASE SQL_COMPANY_DB;
USE SQL_COMPANY_DB;
-- 1. DEPARTMENTS
CREATE TABLE Departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);
INSERT INTO Departments VALUES
(1,'HR'),
(2,'IT'),
(3,'Finance'),
(4,'Sales'),
(5,'Marketing');
-- 2. EMPLOYEES
CREATE TABLE Employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    dept_id INT,
    salary DECIMAL(10,2),
    city VARCHAR(50)
);
INSERT INTO Employees VALUES
(101,'Arun',2,55000,'Chennai'),
(102,'Bala',2,60000,'Madurai'),
(103,'Anitha',1,45000,'Trichy'),
```

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(104,'David',3,NULL,'Coimbatore'),
(105,'Asha',4,70000,'Madurai'),
(106,'John',NULL,50000,'Salem'),
(107,'Kavin',4,80000,'Chennai'),
(108,'Meena',5,65000,'Madurai'),
(109,'Priya',2,60000,'Trichy'),
(110,'Ravi',4,80000,'Madurai');
-- 3. CUSTOMERS
-- Includes duplicates
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    email VARCHAR(100),
    city VARCHAR(50)
);
INSERT INTO Customers VALUES
(1,'Raj','raj@gmail.com','Madurai'),
(2,'Kumar','kumar@gmail.com','Chennai'),
(3,'Divya','divya@gmail.com','Madurai'),
(4,'Raj','raj@gmail.com','Madurai'),
(5,'Sneha','sneha@gmail.com','Trichy'),
(6,'Vijay','vijay@gmail.com','Chennai'),
(7,'Rani','rani@gmail.com','Madurai'),
(8,'Arun','arun@gmail.com','Salem');
-- 4. PRODUCTS
CREATE TABLE Products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(50),
    category VARCHAR(50),
    price DECIMAL(10,2)
);
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INSERT INTO Products VALUES
(201,'Laptop','Electronics',55000),
(202,'Mobile','Electronics',25000),
(203,'Shoes','Fashion',3000),
(204,'Watch','Accessories',5000),
(205,'Tablet','Electronics',35000);
-- 5. ORDERS
CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    product_id INT,
    order_date DATE,
    quantity INT
);
INSERT INTO Orders VALUES
(1001,1,201,'2025-01-10',1),
(1002,2,202,'2025-01-15',2),
(1003,1,203,'2025-02-05',3),
(1004,3,204,'2025-02-12',1),
(1005,4,202,'2025-03-01',1),
(1006,5,201,'2025-03-10',1),
(1007,1,205,'2025-04-05',1),
(1008,6,202,'2025-04-15',2),
(1009,1,204,'2025-05-01',1),
(1010,7,203,'2025-05-12',2),
(1011,1,202,'2025-06-01',1),
(1012,1,203,'2025-06-15',1);
-- 6. SALES
CREATE TABLE Sales (
    sale_id INT PRIMARY KEY,
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    order_date DATE,
    product_id INT,
    amount DECIMAL(10,2)
);
INSERT INTO Sales VALUES
(1,'2025-01-10',201,55000),
(2,'2025-01-15',202,50000),
(3,'2025-02-05',203,9000),
(4,'2025-02-12',204,5000),
(5,'2025-03-01',202,25000),
(6,'2025-03-10',201,55000),
(7,'2025-04-05',205,35000),
(8,'2025-04-15',202,50000),
(9,'2025-05-01',204,5000),
(10,'2025-05-12',203,6000),
(11,'2025-06-01',202,25000),
(12,'2025-06-15',203,3000);
-- 7. MONTHLY SALES
CREATE TABLE MonthlySales (
    month VARCHAR(20),
    sales DECIMAL(10,2)
);
INSERT INTO MonthlySales VALUES
('Jan',105000),
('Feb',14000),
('Mar',80000),
('Apr',85000),
('May',11000),
('Jun',28000);
```

```
-- 8. SALES TEAM
CREATE TABLE SalesTeam (
   emp_name VARCHAR(50),
   region VARCHAR(50),
   sales DECIMAL(10,2)
);
INSERT INTO SalesTeam VALUES
('Arun','South',80000),
('Bala','South',90000),
('Divya','North',70000),
('John','North',95000),
('Meena','East',60000),
('Priya','East',75000),
('Ravi','West',85000),
('Kavin','West',92000);
```

Scenario Questions

1. Find duplicate customer records from the Customers table.
2. Replace NULL salaries with 0 in the Employees table.
3. Fetch the top 5 highest-paid employees.
4. Display employees whose names start with letter **A**.
5. Show all orders placed between two given dates.
6. Show employee names along with department names.
7. Find employees who are not assigned to any department.
8. Show all departments even if no employees are assigned.
9. Find records present in one table but missing in another table.
10. Show customers who placed orders and customers who did not place orders.
11. Calculate total sales month-wise from the Sales table.
12. Find average salary department-wise.
13. Find which department has the maximum number of employees.
14. Find the second highest salary in the Employees table.
15. Find duplicate emails in the Customers table.

16. Compare this month sales with last month sales.
17. Find top 3 products by revenue.
18. Find which city has the highest number of customers.
19. Find customers who purchased more than 5 times.
20. Find inactive customers who have not ordered in the last 6 months.

Window Function Scenarios

21. Rank employees based on salary.
22. Calculate running total of sales by date.
23. Show previous month sales using LAG() function.
24. Find top performer in each sales region.
25. Assign row numbers to duplicate customer records.
26. Create an index to improve employee name search performance.
27. Improve query performance on large sales table using indexing.
28. Use CTE to fetch employees with salary greater than 50,000.
29. Create an index on customer email for faster lookup.
30. Extract today's sales report automatically.
31. Generate total sales report from the Sales table.
32. Remove duplicate customer records keeping one record.
33. Join Customers, Orders, and Products tables.
34. Generate monthly sales report.
35. Find highest-selling product based on number of orders.